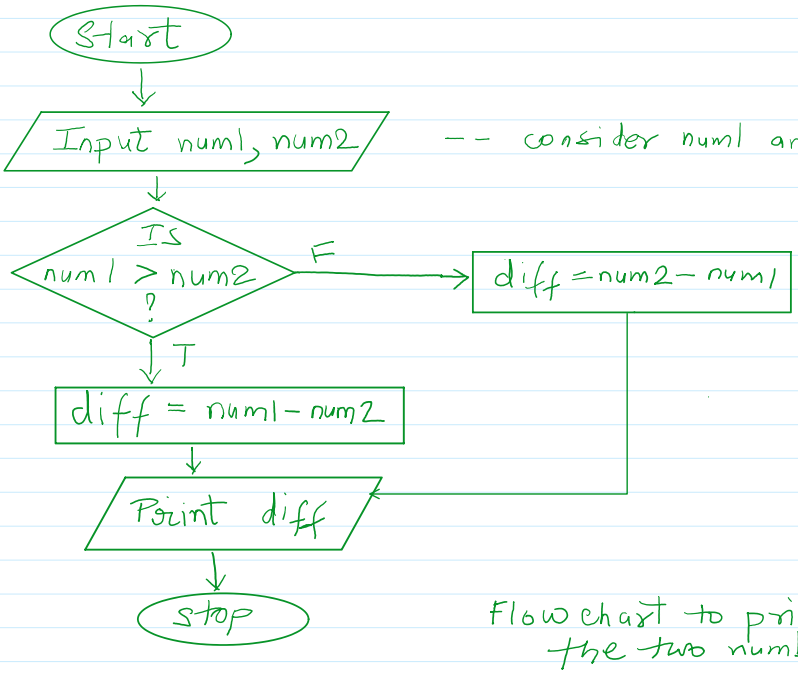
Flow of control

31 August 2024 16:18

Selection

Now suppose we have rupees 10 to buy a pen. on visiting the stationary shop there are varieties of pen priced at Rupees 10 each. Here we have to decide which pen to buy. Similarly, when we use the directional services of a digital map to reach home from one place to another, we notice that it shows more than one path like the least crowded path or short test distance path, Etc. We decide the path as per our priority. A decision involves selecting from one of the two or more possible options. in programming, this concept of decision making or selection is implemented. is implemented with the help of **if… else…** statement.



Flowchart to print positive difference of the two numbers num1 and num2.

The syntax of if statement :

```
if condition :
    statement(s)
```

The syntax of if... else statement:

```
if condition :
    statement(s).
else:
    statement(s).
```

The syntax of if... elif... else statement

```
if condition :
    statement(s)
elif condition :
    statement(s)
elif condition :
    statement(s)
...
else :
    statement(s)
```

WAP to print the positive difference of two numbers.

```
num1 = int(input(" Enter first number: "))
num2 = int(input(" Enter second number: "))

if num1 < 0 or num2 < 0 :
    print ("Error! numbers cannot be negative").
else:
    if num1 > num2 :
        diff = num1 - num2
    else:
        diff = num2 - num1
    print ("The difference of", num1, "and", num2, "is", diff)
```

Write a program to create a simple calculator to perform basic arithmetic operations on two numbers. The program should do the following:

- Accept two numbers from the user.
- Ask user to input any of the operator. (+, -, *, /, //(lower division), %(remainder), ^(exponentiation)). An error message is displayed if the user enters anything else.

Write a program to input three angles of a triangle and check whether its construction is possible or not. If possible, then check and display whether it is an acute angle triangle, right angled or an obtuse angle triangle. Otherwise, display a "triangle is not possible".

```
# Write a program to input three angles of a
# triangle and check whether its construction is
# possible or not. If possible, then check and
# display whether it is an acute angle triangle,
# right angled or an obtuse angle triangle.
# Otherwise, display a "triangle is not possible".

a = float(input('Enter first angle: '))
b = float(input('Enter second angle: '))
c = float(input('Enter third angle: '))
sum = a + b + c
if a == 0 or b == 0 or c == 0 or sum != 180:
    print('Triangle is not possible')
else:
    print('Triangle is possible')
    if a == 90 or b == 90 or c == 90 :
        print('Right Angled Triangle')
    elif a > 90 or b > 90 or c > 90 :
        print('Obtuse Angled Triangle')
    else:
        print('Acute Angled Triangle')
```

Home Work

1. Write a program to input the cost, price and the selling price of an article If the selling price is more than the cost price, then calculate and display the actual profit and the profit percentage, otherwise, calculate and display the actual loss and loss percentage. If the cost price and the selling price are equal, the program displays the message, "neither profit nor loss".
2. Write a program to input three numbers and check whether they are equal or not. If they are unequal numbers, and then display the greatest among them. Otherwise, display the message. All the numbers are equal.

Sample input: 34, 87, 61
Sample output: Greatest number: 87

Sample input: 81, 81, 81
Sample output: All the numbers are equal

3. Write a program to accept a number and check whether the number is divisible by 3 as well as 5. Otherwise, decide:
   a. Is number divisible by 3 and not by 5?
   b. Is the number divisible by 5 and not by 3?
   c. Is the number neither divisible by 3NOR by 5.
   The program displays the message accordingly.

4. Write a program to input the year. The program checks and displays whether it is:
   a. A leap year.
   b. A century Leap year.
   c. First century year, but not a leap year.
   For example:
      INPUT: 1900
      OUTPUT: A century year but not a leap year.
      INPUT: 2000
      OUTPUT: A century leap year.
      INPUT: 2024
      OUTPUT: A leap year.

5. Write a program to input to unequal positive numbers and check whether they are perfect square numbers or not. If the user enters any negative number, then the program display the message **"square root of negative number cannot be determined"**.
   Sample input: 81, 100
   Sample output: They are perfect square numbers.

   Sample input: 225, 99
   Sample output: 225 is a perfect square number.
                 99 is not a perfect square number.

$3^2 = 9$

$\sqrt{9} = 3$

$\sqrt{20} = \sqrt{2 \times 2 \times 5} = 2\sqrt{5}$

$\sqrt{81} = \sqrt{3 \times 3 \times 3 \times 3}$

$= 3 \times 3 = 9$

A variable is a container, in which we keep values.

$x = 5$

$x \rightarrow \boxed{5}$

In programming a variable is a named memory location.

6. Accept three unequal numbers and display the second smallest number. Without using if-else statement.
   For example:
   Input: First Number: 100
          Second Number: 50
          Third Number: 75
   Output: 75

N1	N2	N3
100	50	75

Hint:- Go to google type math library in Python.
→ max() and min() function.

7. A pre-paid taxi charges from passenger as per the tariff given below:

Distance	Rate
Up to 5 KM	₹100
For the next 10 KM	₹20/KM
For the next 10 KM	₹15/KM
More than 25 KM	₹12/KM

DC

20 Km = (100 + 20×10) + 15×(20−15)
= 375.

Write a program to input the distance covered and calculate The amount to be paid by the passenger. The program displays the printed bill with the details given below:

```
Input: Enter Taxi number: 4454
        Distance covered: 20

Bill
--------------------------
Taxi No. : 4454
Distance Covered: 20 KM
Amount: ₹(375)
--------------------------
```

55 Km =
1st 5 Km = 100
2nd 10 Km = 20×10 = 200
3rd 10 Km = 15×10 = 150
Distance left (55−25) = 12×30 = 360
710

Home work:
Flow chart and Program

Q8. For clothes showroom has announced festive discounts and the gifts on the purchase of items based on the total cost as given below:

Total cost	Discount	Gift
Up to ₹2000	5%	Calculator
₹2001 to ₹5000	10%	School Bag
₹5001 to ₹10000	15%	Wall Clock
Above ₹10000	20%	Wrist watch

Write a program to input the total cost. Compute and display the amount to be paid by the customer, along with the gift.
For example:
Input: Bill Amount: ₹3000
Output: Discount 10% : ₹300
Amount to pay: ₹2700
Gift: School Bag

Q9. An employee wants to deposit a certain sum of money under term deposits scheme in a bank. The bank has provided the tariffs of the scheme, which is given below:

No. of days	Rate of Interest
Up to 180	5.5%
181 to 364	7.5%
Exact 365 days	9.0%
More than 365 days	8.5%

Write a program to calculate and display the maturity amount, taking the sum and the number of days as Inputs.
For example:
Input: Number of days: 365

Amount to Invest: ₹1000

Output:

Maturity amount: ₹1090

Q10 Mr. Kumar is an LIC agent. He offers discount to his policy holders on the annual premium. However, he also gets commission on the sum assured as well, the given tariff.

Sum Assured	Discount	Commission
Up to ₹100000	5%	2%
₹100001 to ₹200000	8%	3%
₹200001 to ₹500000	10%	5%
More than ₹500000	15%	7.5%

American → ASCII

Write a program to input the name of the policyholder sum assured and the first annual premium. Calculate the discount of the policy holder and the commission of the agent. The program displays all the details as (premium for one month = (sum assured)/12:

Unicode value

'\u 20b9'

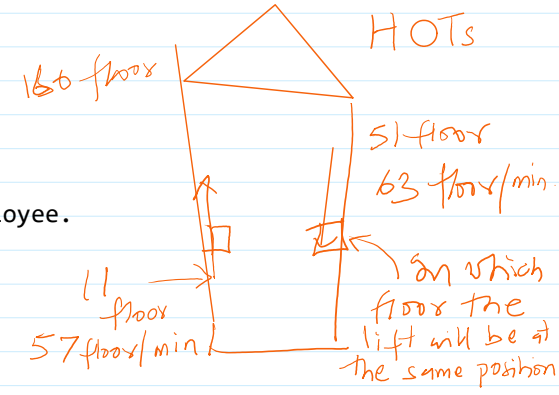
Input:
Name of the policyholder: Ram Singh
Sum assured: ₹250000

Output:
Discount on the first premium is 10%: ₹2083.3333
Commission of the agent 5%: ₹1041.6667

Calculation of discount value:
(250000/12)*0.1 = 2083.3333
(250000/12)*0.05 = 1041.6667

Q11. A company announces revised DNS allowance and special allowances for their employees as per tariffs given below:

Basic	Dearness Allowance(DA)	Special Allowance(SA)
Up to ₹10000	10%	5%
₹10001 to ₹20000	12%	8%
₹20001 to ₹30000	15%	10%
₹30001 and above	20%	12%



Write a program to accept a name and basic salary (BS) Of an employee. Calculate and display gross salary(GS).
GS = BS + DA + SA

Input:
Name: Eshan Kumar
BASIC SALARY:

Output:
DA:
SA:
GS: